

Notes

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Logarithm

Domain

$(0, \infty)$

Range

$(-\infty, \infty)$

Polynomials

Factor theorem

The factor theorem states that $(x - r)$ is a factor of a polynomial $p(x)$ if and only if the root $x = r$ is also a root of $p(x)$.

Exercises

- The polynomial $-5x^4 + 8x^3 + 29x^2 - 20x - 12$ can be expressed as $-(x - 1)(x - 3) \cdot P(x)$. Find $P(x)$.

First let's divide by $(x - 1)$

$$\begin{array}{r|rrrrrr} 1 & -5 & 8 & 29 & -20 & -12 \\ & & -5 & 3 & 32 & 12 \\ \hline & -5 & 3 & 32 & 12 & 0 \end{array}$$

So we have $(x - 1) \cdot (-5x^3 + 3x^2 + 32x + 12)$

Let's divide by $(x - 3)$

$$\begin{array}{r|rrrr} 3 & -5 & 3 & 32 & 12 \\ & & -15 & -36 & -12 \\ \hline & -5 & -12 & -4 & 0 \end{array}$$

So we have $(x - 1)(x - 3)(-5x^2 - 12x - 4)$

So we have $(x - 1) \cdot (x - 3) \cdot (-5x^2 - 2x + 4)$

Finally, $-(x - 1)(x - 3)(5x^2 + 2x + 4)$. So $P(x) = 5x^2 + 2x + 4$.

Factoring further, $P(x) = (5x + 2)(x + 2)$.

- Find the polynomial $P(x)$ given that

$$4x^4 - 19x^3 - 9x^2 + 19x + 5 = (x - 1)(x - 5)P(x)$$

First let's divide by $(x - 1)$.

$$\begin{array}{r|rrrrrr} 1 & 4 & -19 & -9 & 19 & 5 \\ & & 4 & -15 & -24 & -5 \\ \hline & 4 & -15 & -24 & -5 & 0 \end{array}$$

So $(x - 1)(4x^3 - 15x^2 - 24x - 5)$. Now divide by $(x - 5)$.

$$\begin{array}{r|rrrr} 5 & 4 & -15 & -24 & -5 \\ & & 20 & 25 & 5 \\ \hline & 4 & 5 & 1 & 0 \end{array}$$

So $(x - 1)(x - 5)(4x^2 + 5x + 1)$.

- Find $P(x)$ given $4x^4 - 8x^3 - 3x^2 + 7x - 2 = (x - 2)(x + 1)P(x)$.

First divide by $(x + 1)$:

$$\begin{array}{r|rrrrr}
 -1 & 4 & -8 & -3 & 7 & -2 \\
 & & -4 & 12 & -9 & 2 \\
 & 4 & -12 & 9 & -2 & 0
 \end{array}$$

So $(x+1)(4x^3 - 12x^2 + 9x - 2)$. Now divide the remainder by $(x-2)$:

$$\begin{array}{r|rrrr}
 2 & 4 & -12 & 9 & -2 \\
 & & 8 & -8 & 2 \\
 & 4 & -4 & 1 & 0
 \end{array}$$

So $(x+1)(x-2)(4x^2 - 4x + 1)$. So $P(x) = (4x^2 - 4x + 1)$. Factoring further,

$$\begin{aligned}
 P(x) &= 4x^2 - 4x + 1 \\
 &= (4x^2 - 2x) + (-2x + 1) \\
 &= 2x(2x - 1) - 1(2x - 1) \\
 &= (2x - 1)^2
 \end{aligned}$$

- Let $p(x) = -6x^4 + 5x^3 + 52x^2 - 19x - 60$. Given that $p(-1) = p(3) = 0$, find the sum of all distinct roots of $p(x)$.

$p(-1) = 0$ means $x = -1$ is a root, so $(x+1)$ is a factor.

$p(3) = 0$ means $x = 3$ is a root, so $(x-3)$ is a factor.

Let's first divide by $(x+1)$.

$$\begin{array}{r|rrrrr}
 -1 & -6 & 5 & 52 & -19 & -60 \\
 & & 6 & -11 & -41 & 60 \\
 & -6 & 11 & 41 & -60 & 0
 \end{array}$$

So $p(x) = (x+1)(-6x^3 + 11x^2 + 41x - 60)$. Now let's divide the remainder by $(x-3)$.

$$\begin{array}{r|rrrr}
 3 & -6 & 11 & 41 & -60 \\
 & & -18 & -21 & 60 \\
 & -6 & -7 & 20 & 0
 \end{array}$$

So $p(x) = (x+1)(x-3)(-6x^2 - 7x + 20)$.

Factoring the remainder further:

$$\begin{aligned}
 p(x) &= (x+1)(x-3)(-6x^2 - 7x + 20) \\
 &= (x+1)(x-3)(-6x^2 + 8x) + (-15x + 20) \\
 &= (x+1)(x-3)(2x+5)(-3x+4)
 \end{aligned}$$

The roots are $x = -1, 3, -\frac{5}{2}, \frac{4}{3}$.

The number we are looking for is $-1 + 3 - \frac{5}{2} + \frac{4}{3} = 2 - \frac{5}{2} + \frac{4}{3} = -\frac{1}{2} + \frac{4}{3} = \frac{5}{6}$.

- Find the distinct roots of $x^4 - 5x^2 + 4$ given $p(1) = p(-1) = 0$.

$(x-1)$ is a factor, let's divide by it first:

$$\begin{array}{r|rrrrr}
 1 & 1 & 0 & -5 & 0 & 4 \\
 & & 1 & 1 & -4 & -4 \\
 & 1 & 1 & -4 & -4 & 0
 \end{array}$$

So $(x-1)(x^3 + x^2 - 4x - 4)$. Now divide by $(x+1)$.

$$\begin{array}{r|rrrr}
 -1 & 1 & 1 & -4 & -4 \\
 & & -1 & 0 & 4 \\
 & 1 & 0 & -4 & 0
 \end{array}$$

So $(x-1)(x+1)(x^2-4) = (x-1)(x+1)(x+2)(x-2)$.

- Find $P(x)$ given $x^4 - 2x^3 - 7x^2 + 18x - 18 = (x-3)(x+3)P(x)$.

First let's divide by $(x-3)$:

$$\begin{array}{r|rrrrr}
 3 & 1 & -2 & -7 & 18 & -18 \\
 & & 3 & 3 & -12 & 18 \\
 & 1 & 1 & -4 & 6 & 0
 \end{array}$$

So the remainder is $x^3 + x^2 - 4x + 6$. Let's divide by $(x+3)$.

$$\begin{array}{r|rrrr}
 -3 & 1 & 1 & -4 & 6 \\
 & & -3 & 6 & -6 \\
 & 1 & -2 & 2 & 0
 \end{array}$$

We have $P(x) = x^2 - 2x + 2$.

- Find $P(x)$ given that $x^4 + x^3 - 36x^2 + 37x - 35 = (x-5)(x+7)P(x)$.

First divide by $(x-5)$:

$$\begin{array}{r|rrrrr}
 5 & 1 & 1 & -36 & 37 & -35 \\
 & & 5 & 30 & -30 & 35 \\
 & 1 & 6 & -6 & 7 & 0
 \end{array}$$

Remainder: $x^3 + 6x^2 - 6x + 7$. Now divide that by $(x+7)$:

$$\begin{array}{r|rrrr}
 -7 & 1 & 6 & -6 & 7 \\
 & & -7 & 7 & -7 \\
 & 1 & -1 & 1 & 0
 \end{array}$$

Result: $x^2 - x + 1$

Polynomial long division

Exercises

- What is the remainder of $3x^2 - 2x^2 - 18x - 1$ divided by $3x^2 - 8x - 2$?

$$\begin{array}{r}
 \begin{array}{r}
 \times + 2 \\
 3x^2 - 8x - 2 \mid \begin{array}{rrrr}
 3 & -2 & -18 & -1 \\
 3 & -8 & -2 & \\
 \hline
 & 6 & -16 & \\
 & 6 & -16 & -4 \\
 & & & 3
 \end{array}
 \end{array}
 \end{array}$$

The remainder is 3.

- What is the remainder when $x^4 - 2x^3 - 4x^2 - 10x - 7$ is divided by $x^2 + 2x + 1$?

$$\begin{array}{r}
 \begin{array}{r}
 \times^2 - 4x + 3 \\
 x^2 + 2x + 1 \mid \begin{array}{rrrrr}
 1 & -2 & -4 & -10 & -7 \\
 1 & 2 & 1 & & \\
 \hline
 & -4 & -5 & & \\
 & -4 & -8 & -4 & \\
 & & & &
 \end{array}
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 3 \quad -6 \\
 3 \quad 6 \quad 3 \\
 \hline
 -12x \quad -10
 \end{array}$$

The remainder is $-12x - 10$.

- What is the quotient of $2x^4 - x^2 - 1$ divided by $2x^2 + 1$?

$$\begin{array}{r}
 x^2 \quad -1 \\
 2x^2 + 1 \mid 2 \quad 0 \quad -1 \quad 0 \quad -1 \\
 2 \quad 0 \quad 1 \\
 \hline
 -2 \\
 -2 \quad 0 \quad 1 \\
 \hline
 0
 \end{array}$$

So it's $x^2 - 1$.

Rational roots theorem

The rational roots theorem (a.k.a. rational zeros theorem) gives the rational roots of a polynomial with integer coefficients.

Theorem. Suppose that $\frac{p}{q} \in \mathbb{Q}$, in lowest terms, is a root of a polynomial with integer coefficients. Then p must be a factor of the polynomial's constant term, and q must be a factor of its leading coefficient.

That means every root must take the form $\pm \frac{\text{factor of the constant term}}{\text{factor of the leading coefficient}}$.

Exercises

- For the polynomial $f(x) = 3x^8 - 18x^2 - 33x - 18$, how many potential roots are given by the rational roots theorem?

The numerator must be a factor of -18 . The denominator must be a factor of 3 . So here are the possibilities:

$$\pm \frac{1}{3}, \pm \frac{2}{3}, \pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18$$

So 16.

- Given that the polynomial $2x^3 + x^2 + 5x - 3$ has a root in the interval $(0, \frac{3}{2})$, find the sum of the distinct real roots of $f(x)$.

TODO

Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$\begin{aligned}
& \frac{1}{1 + \sqrt{2}} \\
&= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\
&= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\
&= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\
&= \frac{1 - \sqrt{2}}{1 - 2} \\
&= \frac{1 - \sqrt{2}}{-1} \\
&= -1 + \sqrt{2}
\end{aligned}$$

Exercises

•

$$\begin{aligned}
& \frac{1}{1 + \sqrt{2}} \\
&= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\
&= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\
&= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\
&= \frac{1 - \sqrt{2}}{1 - 2} \\
&= \frac{1 - \sqrt{2}}{-1} \\
&= -1 + \sqrt{2}
\end{aligned}$$

1.

$$\begin{aligned}
& \frac{1 + \sqrt{8}}{\sqrt{9} - \sqrt{2}} \\
&= \frac{(1 + \sqrt{8})(\sqrt{9} + \sqrt{2})}{7} \\
&= \frac{\sqrt{9} + 3\sqrt{8} + \sqrt{2} + \sqrt{16}}{7} \\
&= \frac{3 + 6\sqrt{2} + \sqrt{2} + 4}{7} \\
&= \frac{7 + 7\sqrt{2}}{7} \\
&= 1 + \sqrt{2}
\end{aligned}$$

2.

$$\begin{aligned}
\frac{x^2 y^2}{\sqrt{x^3 y^3}} &= \frac{x^2 y^2 \sqrt{x^3 y^3}}{x^3 y^3} \\
&= \frac{x^2 y^2 x^{3/2} y^{3/2}}{x^3 y^3} \\
&= \frac{x^{7/2} y^{7/2}}{x^3 y^3} \\
&= \sqrt{xy}
\end{aligned}$$

3.

$$\begin{aligned}
\frac{\sqrt{12x}}{3\sqrt{2x^2y}} &= \frac{\sqrt{12x} \cdot \sqrt{2x^2y}}{3 \cdot 2x^2y} \\
&= \frac{\sqrt{24x^3y}}{6x^2y} \\
&= \frac{\sqrt{4x^2 \cdot 6xy}}{6x^2y} \\
&= \frac{2x \cdot \sqrt{6xy}}{6x^2y} \\
&= \frac{\sqrt{6xy}}{3xy}
\end{aligned}$$

4. Given that $d > 0$.

$$\begin{aligned}
 \frac{cd}{\sqrt{cd^3}} &= \frac{cd\sqrt{cd^3}}{cd^3} \\
 &= \frac{\sqrt{cd^3}}{d^2} \\
 &= \frac{\sqrt{c} \cdot d^{3/2}}{d^2} \\
 &= \frac{\sqrt{c} \cdot d^{1/2}}{d} \\
 &= \frac{\sqrt{cd}}{d}
 \end{aligned}$$

5. Given that $p > 0$

$$\begin{aligned}
 \frac{7t}{\sqrt{14tp^2}} &= \frac{7t \cdot \sqrt{14tp^2}}{14tp^2} \\
 &= \frac{7tp\sqrt{14t}}{14tp^2} \\
 &= \frac{\sqrt{14t}}{2p}
 \end{aligned}$$

6. Given that $a > 0$

$$\begin{aligned}
 \sqrt{\frac{5b}{a^3}} &= \frac{\sqrt{5b}}{\sqrt{a^3}} \\
 &= \frac{\sqrt{5b}}{a\sqrt{a}} \\
 &= \frac{\sqrt{5ab}}{a\sqrt{a}^2} \\
 &= \frac{\sqrt{5ab}}{a^2}
 \end{aligned}$$

7.

$$\begin{aligned}
 \sqrt{\frac{14r}{p^5}} &= \frac{\sqrt{14r}}{\sqrt{p^5}} \\
 &= \frac{\sqrt{14p^5r}}{p^5} \\
 &= \frac{\sqrt{14r} \cdot p^{5/2}}{p^5} \\
 &= \frac{\sqrt{14pr}}{p^3}
 \end{aligned}$$

8.

$$\begin{aligned}
 \sqrt{\frac{9a^2}{b^3}} &= \frac{\sqrt{9a^2}}{\sqrt{b^3}} \\
 &= \frac{\sqrt{9a^2}}{b\sqrt{b}} \\
 &= \frac{\sqrt{9a^2b}}{b\sqrt{b}^2} \\
 &= \frac{3a\sqrt{b}}{b^2}
 \end{aligned}$$

9. Given that $z > 0$.

$$\begin{aligned}
 \sqrt{\frac{11}{wz^2}} &= \frac{\sqrt{11}}{\sqrt{wz^2}} \zeta = \frac{\sqrt{11}}{z\sqrt{w}} \\
 &= \frac{\sqrt{11w}}{zw}
 \end{aligned}$$

10.

$$\begin{aligned}
 \sqrt{\frac{5a}{3b}} &= \frac{\sqrt{5a}}{\sqrt{3b}} \\
 &= \frac{\sqrt{15ab}}{3b}
 \end{aligned}$$

11.

$$\begin{aligned}
 \sqrt{\frac{a^2+3}{a^2-3}} &= \frac{\sqrt{a^2+3}}{\sqrt{a^2-3}} \\
 &= \frac{\sqrt{(a^2+3)(a^2-3)}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3} \\
 &= \frac{\sqrt{a^4-9}}{a^2-3}
 \end{aligned}$$

12. Given $y > 0$

$$\begin{aligned}
 \frac{2+3\sqrt{xy}}{3\sqrt{y}} &= \frac{(2+3\sqrt{xy}) \cdot \sqrt{y}}{3y} \\
 &= \frac{3y\sqrt{x}+2\sqrt{y}}{3y} \\
 &= \sqrt{x} + \frac{2\sqrt{y}}{3y}
 \end{aligned}$$

Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x-1)(3x-2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x-1)(3x-2)} = \frac{A}{x-1} + \frac{B}{3x-2}$$

Multiplying both sides by $(x-1)(3x-2)$, we get:

$$1 = A(3x-2) + B(x-1)$$

Setting $x = 1$, we get:

$$1 = A(3(1)-2) + B(1-1) = A$$

Setting $x = \frac{2}{3}$, we get:

$$1 = A\left(3\left(\frac{2}{3}\right) - 2\right) + B\left(\frac{2}{3} - 1\right) = -\frac{B}{3}$$

Solving for B , we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

Exercises

1. Decompose $\frac{44x+48}{(x+6)(x-3)(x+2)}$

$$\frac{44x+48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$

$$44x+48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set $x = 3$:

$$44 \cdot 3 + 48 = B \cdot 9 \cdot 5$$

$$\frac{180}{9 \cdot 5} = B$$

$$B = 4$$

Set $x = -2$:

$$44 \cdot -2 + 48 = C \cdot 4 \cdot -5$$

$$-40 = C \cdot -20$$

$$C = 2$$

Set $x = -6$:

$$44 \cdot -6 + 48 = A \cdot -9 \cdot -4$$

$$A = -6$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find B

$$\begin{aligned}\frac{-2x-9}{x+3}x &= \frac{1}{x+3} + \frac{B}{x} \\ (-2x-9) &= x + B(x+3) \\ -3x-9 &= B(x+3) \\ \frac{-3(x+3)}{x+3} &= B \\ B &= -3\end{aligned}$$

3. Given that

$$\begin{aligned}\frac{-9x+36}{(x-3)(x-6)} &= -\frac{3}{x-3} + \frac{B}{x-6} \\ -9x+36 &= -3(x-6) + B(x-3) \\ -6x+18 &= B(x-3) \\ \frac{-6(x-3)}{x-3} &= B \\ B &= -6\end{aligned}$$

Functions

Horizontal asymptotes of rational functions

Rational functions are of the form:

$$f(x) = \frac{ax^n + bx^{n-1} + \dots}{cx^m + dx^{m-1} + \dots}$$

To get the horizontal asymptotes, you first take the **dominant term**. It's the leading term of the polynomial in the denominator, *without the coefficient*. Here x^m .

Then, you divide every term in both numerator and denominator, and evaluate at $x \rightarrow \infty$.

Radical functions

Domain

All the values where the radicand > 0 for even degrees, $(-\infty, \infty)$ for odd degrees.

Exercises

- Domain of $d(x) = \sqrt[7]{-2x + \frac{1}{2}} + 1$?

$$-2x + \frac{1}{2} >$$

Roots of rational functions

First factor numerator and denominator, then cancel out common factors, then set the numerator to zero and solve.

Exercises

- Solve $f(t) = 0$ where $f(t) = \frac{7t^2 + 14}{t^2 - 25}$.

$$\frac{7t^2 + 14}{t^2 - 25} = 0$$

$$\frac{7(t^2 + 2)}{(t + 5)(t - 5)} = 0$$

$$t^2 + 2 = 0$$

$$t^2 = -2$$

2. Solve $f(x) = 0$ where $f(x) = \frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45}$.

$$\frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45} = 0$$

$$\frac{x^2 + 5x - 3x - 15}{x^2(x + 5) - 9(x + 5)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x^2 - 9)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x - 3)(x + 3)} = 0$$

$$1 = 0$$

The equation has no solution.

3. Solve $f(t) = 0$ where $f(t) = \frac{t-3}{t^2+6t+5}$

$$\frac{t-3}{t^2+6t+5} = 0$$

$$\frac{t-3}{t^2+t+5t+5} = 0$$

$$\frac{t-3}{(t+5)(t+1)} = 0$$

$$t-3 = 0$$

$$t = 3$$

Invertible functions

A function is invertible on an interval if and only if it is one-to-one on that interval.

To check if a function is one-to-one, we can use the horizontal line test. If any horizontal line intersects the graph of the function at most once, then the function is one-to-one.

Statistics

Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Covariance

Covariance is the measure of the joint variability of two random variables.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Sum of squares

The sum of squares is the sum of the squared differences between each data point and the mean.

$$\sum_{i=1}^n (x_i - \bar{x})^2$$

Denoted S_{xx} . If two variables are involved (covariance), it is denoted as S_{xy} .

Shortcut formula:

$$S_{xx} = \sum_{i=1}^n x_i^2 - \frac{1}{n \left(\sum_{i=1}^n x_i \right)^2}$$
$$S_{xy} = \sum_{i=1}^n x_i y_i - \frac{1}{n \left(\sum_{i=1}^n x_i \right)} \left(\sum_{i=1}^n y_i \right)$$

Mean estimation for continuous grouped data

$$\bar{x} \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot m_i$$

where

- f_i is the frequency of the i th interval
- m_i is the midpoint of the i th interval
- n is the total number of observations

Estimating a variance for continuous grouped data

Using the same notation as above:

$$\sigma_n^2 \approx \frac{1}{n} \sum_{i=1}^K f_i \cdot (m_i - \bar{x})^2$$

Graphing

Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

Example

Given that $(x - 3)$ is a factor of $f(x) = x^3 + x^2 - 4x - 24$, sketch the curve of $y = f(x)$.

First, synthetic division:

	x^3	x^2	x^1	x^0
3	1	1	-4	-24
		3	12	24
	1	4	8	0

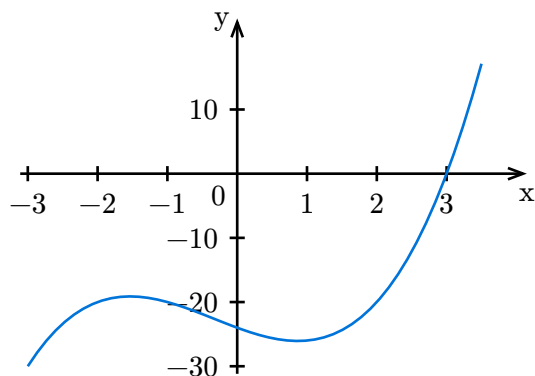
So we have $f(x) = (x - 3)(x^2 + 4x + 8)$

$\therefore x = 3$ is a root.

The discriminant of $x^2 + 4x + 8$ is $4^2 - 4 \cdot 1 \cdot 8 = -16$, which is negative, so $x = 3$ is the only real root.

The y intercept is $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



Logarithms

Domain

$(0, \infty)$

Range

$(-\infty, \infty)$

Complex Numbers

Magnitude

The magnitude of a complex number $z = a + bi$ is defined as $|z| = \sqrt{a^2 + b^2}$.

Argument

For any complex number, like $z = 3 + 4i$, there is $\arg(z)$ that is the angle (in radians) between the x -axis and the complex number in the Argand diagram.

So for z :

$$\begin{aligned}\tan \theta &= \frac{4}{3} \\ \theta &= \arctan\left(\frac{4}{3}\right) \\ &\approx 0.93\end{aligned}$$

Trigonometry

Unit circle

Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the x -axis. It is always a positive angle less than or equal to 90 degrees.

Pythagorean identity in the first quadrant

In the first quadrant, any point (x, y) forms a right triangle with the origin, where x and y are the lengths of the legs and $\sqrt{x^2 + y^2}$ is the length of the hypotenuse. Since in the unit circle, the length of the hypotenuse is 1, by definition, the Pythagorean identity states that $x^2 + y^2 = 1$.

Exercises

1. For a point in the first quadrant with $y = \frac{\sqrt{5}}{3}$, find its x coordinate.

$$\sqrt{\left(\frac{\sqrt{5}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{5}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{5}}{3}\right)^2$$

$$x^2 = 1 - \frac{5}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \sqrt{\frac{4}{9}}$$

$$x = \frac{2}{3}$$

2. Point in the first quadrant. y coordinate is $\frac{\sqrt{2}}{3}$. What is the x coordinate?

$$\sqrt{\left(\frac{\sqrt{2}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{2}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{2}}{3}\right)^2$$

$$x^2 = 1 - \frac{2}{9}$$

$$x^2 = \frac{7}{9}$$

$$x = \sqrt{\frac{7}{9}}$$

$$x = \frac{\sqrt{7}}{3}$$

Pythagorean trigonometric identity

In the unit circle, since for any central angle $x = \cos \theta$ and $y = \sin \theta$.

That means

$$x^2 + y^2 = 1$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This is the **Pythagorean trigonometric identity**. It works in all quadrants.

Exercises

1. Point P on the unit circle with $x = \frac{2}{5}$. Find $\sin \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \frac{4}{25}}$$

$$\sin \theta = \sqrt{\frac{21}{25}}$$

$$\sin \theta = \frac{\sqrt{21}}{5}$$

2. y coordinate is $\frac{4}{9}$. Find $\cos \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{4}{9}\right)^2 + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \left(\frac{4}{9}\right)^2$$

$$\cos^2 \theta = 1 - \frac{16}{81}$$

$$\cos^2 \theta = \frac{65}{81}$$

$$\cos \theta = \sqrt{\frac{65}{81}}$$

$$\cos \theta = \frac{\sqrt{65}}{9}$$

3. x coordinate is $\frac{\sqrt{7}}{4}$. Find $\csc \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\sin \theta = \sqrt{1 - \frac{7}{16}}$$

$$\sin \theta = \sqrt{\frac{9}{16}}$$

$$\sin \theta = \frac{3}{4}$$

$$\therefore$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{4}{3}$$

4. x coordinate is $\frac{3}{7}$. Find $\csc \theta$.

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \frac{9}{49}}$$

$$\sin \theta = \sqrt{\frac{40}{49}}$$

$$\sin \theta = \frac{\sqrt{40}}{7}$$

$$\therefore$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{7}{\sqrt{40}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7\sqrt{10}}{20}$$

5. y coordinate is $\frac{1}{9}$. Find $\tan \theta$.

We have $\sin \theta = \frac{1}{9}$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{1}{9}\right)^2}$$

$$\cos \theta = \sqrt{1 - \frac{1}{81}}$$

$$\cos \theta = \sqrt{\frac{80}{81}}$$

$$\cos \theta = \frac{\sqrt{80}}{9}$$

$$\cos \theta = \frac{4\sqrt{5}}{9}$$

$\therefore$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{\frac{1}{9}}{\frac{4\sqrt{5}}{9}}$$

$$\tan \theta = \frac{1}{9} \cdot \frac{9}{4\sqrt{5}}$$

$$\tan \theta = \frac{\sqrt{5}}{20}$$

6. x coordinate is $\frac{\sqrt{11}}{6}$. Find $\cot(\theta)$

$$\text{We have } \cos \theta = \frac{\sqrt{11}}{6} \text{ and } \cot(\theta) = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\frac{\sqrt{11}}{6}}{\sin \theta}$$

$$\begin{aligned}
\sin^2 \theta + \cos^2 \theta &= 1 \\
\sin^2 \theta &= 1 - \cos^2 \theta \\
\sin^2 \theta &= 1 - \left(\frac{\sqrt{11}}{6} \right)^2 \\
\sin^2 \theta &= 1 - \frac{11}{36} \\
\sin^2 \theta &= \frac{25}{36} \\
\sin \theta &= \sqrt{\frac{25}{36}} \\
\sin \theta &= \frac{5}{6} \\
&\therefore \\
\cot \theta &= \frac{\cos \theta}{\sin \theta} \\
\cot \theta &= \frac{\frac{\sqrt{11}}{6}}{\frac{5}{6}} \\
\cot \theta &= \frac{\sqrt{11}}{5}
\end{aligned}$$

Quadrantal angles

Quadrantal angles are angles that are multiples of 90° . For example, 0° , 90° , 180° , 270° , and 360° are all quadrantal angles. They lie on an axis.

They are useful to find the values of trigonometric functions. Just remember that for points on the unit circle, $\cos \theta = x$ and $\sin \theta = y$.

The corresponding angles on $[0, 360^\circ]$ are $0, \frac{\pi}{2}, \pi, 3\frac{\pi}{2}, 2\pi$.

Example: $\cos 0 + \sin\left(\frac{\pi}{2}\right) = 1 + 1 = 2$

Examples

- Find the value of $\tan(180^\circ) + \cot(270^\circ)$

$$\begin{aligned}
\tan(180^\circ) + \cot(270^\circ) &= \frac{\sin(180^\circ)}{\cos(180^\circ)} + \frac{\cos(270^\circ)}{\sin(270^\circ)} \\
&= \frac{0}{-1} + \frac{0}{-1} \\
&= 0
\end{aligned}$$

- $\sec\left(\frac{\pi}{2}\right)$?

$$\begin{aligned}
\sec\left(\frac{\pi}{2}\right) &= \frac{1}{\cos\left(\frac{\pi}{2}\right)} \\
&= \frac{1}{0}
\end{aligned}$$

So it's undefined.

Law of sines

In any triangle, where a , b , and c are the lengths of the sides opposite angles A , B , and C respectively, the law of sines states that:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

For acute angles, the formula can be flipped:

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

For obtuse angles, the same formula works, but it will always give acute angles. To find an obtuse angle, subtract the acute angle from 180° .

Exercises

- Find a if $A = 70^\circ$, $B = 35^\circ$, $b = 12$.

$$\begin{aligned}\frac{a}{\sin(A)} &= \frac{b}{\sin(B)} \\ \frac{a}{\sin(70^\circ)} &= \frac{12}{\sin(35^\circ)} \\ a &= 12 \cdot \frac{\sin(70^\circ)}{\sin(35^\circ)} \\ a &\approx 12 \cdot \frac{0.9397}{0.5736} \\ a &\approx 19.66\end{aligned}$$

- R , Q and P form a triangle. $PR = 21m$. $\angle Q = 89^\circ$, $\angle P = 38^\circ$. What is QR ?

$$\begin{aligned}\frac{QR}{\sin(\angle P)} &= \frac{PR}{\sin(\angle Q)} \\ \frac{QR}{\sin(38^\circ)} &= \frac{21}{\sin(89^\circ)} \\ QR &= \sin(38^\circ) \cdot \frac{21}{\sin(89^\circ)}\end{aligned}$$

- $\triangle XYZ$. $\angle Y = 57^\circ$, $XZ = 35$, $XY = 24$. Find $\angle Z$.

$$\begin{aligned}\frac{\sin(\angle Z)}{XY} &= \frac{\sin(\angle Y)}{XZ} \\ \frac{\sin(\angle Z)}{24} &= \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &= 24 \cdot \frac{\sin(57^\circ)}{35} \\ \sin(\angle Z) &\approx 0.575 \\ \angle Z &\approx 35^\circ\end{aligned}$$

Calculus

Limits

Limits of logarithmic functions

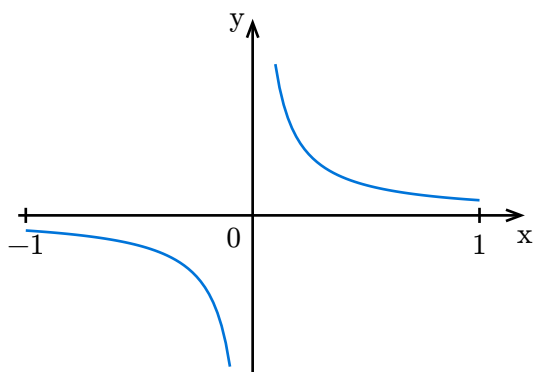
For any real n :

$$\lim_{x \rightarrow n} \log(x) = \log(x)$$

For infinity, remember the domain and apply transformations.

Limits of reciprocal functions

Remember, reciprocal functions have the form $f(x) = \frac{1}{x}$. The limit of a reciprocal function as x approaches infinity is always zero.



In general, for any non-constant polynomial $P(x)$, we have $\lim_{x \rightarrow \pm\infty} \frac{1}{P(x)} = 0$

But for $\lim(x \rightarrow 0)$, we have to consider the degree of the leading term of $P(x)$.

- Any function of the form $y = \frac{1}{x^n}$ where n is odd has the same shape as $y = \frac{1}{x}$. There is no limit at $x = 0$ because left and right limit are not the same.
- Any function of the form $y = \frac{1}{x^n}$ where n is even has the same shape as $y = \frac{1}{x^2}$. There is a limit, since the function will approach ∞ or $-\infty$.

Differentiation

Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x))$$

Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for $y = x^2 + x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 + x) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x) \\ &= 2x + 1 \end{aligned}$$

The power rule

Given a monomial $f(x) = x^n$, where n is a real number,

$$f'(x) = nx^{n-1}$$

Exercises

1. Calculate $\frac{dy}{dx}$ for $y = 2x - \sqrt{x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(2x - \sqrt{x}) \\ &= \frac{d}{dx}2x - \frac{d}{dx}\sqrt{x} \\ &= 2 - \frac{d}{dx}x^{1/2} \\ &= 2 - \frac{1}{2} \cdot x^{-1/2} \\ &= 2 - \frac{1}{2\sqrt{x}}\end{aligned}$$

2. Find the derivative of $f(x) = \frac{1}{8}x^8$

$$\begin{aligned}f'(x) &= \frac{1}{8} \cdot \frac{d}{dx}x^8 \\ &= \frac{1}{8} \cdot 8x^7 \\ &= \frac{8x^7}{8} \\ &= x^7\end{aligned}$$

3. Derivative of $f(x) = 8\sqrt{x}$

$$\begin{aligned}f'(x) &= \frac{d}{dx}8\sqrt{x} \\ &= \frac{d}{dx}8x^{1/2} \\ &= 8 \cdot \frac{d}{dx}\left(\frac{1}{2} \cdot x^{-1/2}\right) \\ &= 8 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{4}{\sqrt{x}}\end{aligned}$$

Rate of change

Average rate of change on an interval $[a, a + h]$:

$$\frac{\delta y}{\delta x} = \frac{f(a + h) - f(a)}{h}$$

The *instantaneous rate of change* at a point $x = a$ is given by:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Exercises

1. Given that $f(x) = x^2$, what is $f'(2)$ according to the definition?

$$\begin{aligned}
 f'(2) &= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (4 + h) \\
 &= 4
 \end{aligned}$$

Integration

The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of x^2 is $2x$, we have x^2 as an antiderivative of $2x$. But $x^2 + 2$, $x^2 + 1$, etc. are also antiderivatives of $2x$.

In general, the antiderivatives of $2x$ follow the pattern $x^2 + C$ where C is a constant.

We can also formulate that using the $\int$ symbol for integration:

$$\int 2x \, dx = x^2 + C$$

Where C is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here $2x$) is called the **integrand**.

The power rule for integration

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

Exercises

1. Antiderivative of $\frac{1}{x^4}$?

$$\begin{aligned}
 \int \frac{1}{x^4} dx &= \int x^{-4} dx \\
 &= \frac{x^{-4+1}}{-4+1} + C \\
 &= \frac{x^{-3}}{-3} + C \\
 &= -\frac{1}{3}x^3 + C
 \end{aligned}$$

2. Antiderivative of $x^{\frac{5}{8}}$

$$\begin{aligned}
 \int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\
 &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\
 &= \frac{8x^{\frac{13}{8}}}{13} + C
 \end{aligned}$$

3. Antiderivative of $\sqrt[5]{x^7}$

$$\begin{aligned}
 \int x^{\frac{7}{5}} dx &= \frac{x^{7/5+1}}{7/5+1} + C \\
 &= \frac{x^{12/5}}{12/5} \\
 &= \frac{5x^{12/5}}{12}
 \end{aligned}$$

4. Calculate $\int \left(\frac{z}{5} dz\right)$

$$\begin{aligned}
 \int \left(\frac{z}{5} dz\right) &= \frac{1}{5} \int z dz \\
 &= \frac{1}{5} \cdot \left(\frac{z^2}{2} + C\right) \\
 &= \frac{z^2}{10} + C
 \end{aligned}$$

The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like $5x^2$, we can take the constant factor out of the integral:

$$\int 5x^2 dx = 5 \cdot \int x^2 dx$$

This is the **constant factor rule**. In general:

$$\int kf(x) dx = k \int f(x) dx$$

Where k is a **constant**. It does not work if k is a variable.

Exercises

1. Calculate $\int \frac{1}{2y^5} dy$

$$\begin{aligned}
 \int \frac{1}{2y^5} dy &= \frac{1}{2} \int y^{-5} \\
 &= \frac{1}{2} \cdot \frac{y^{-5+1}}{-5+1} + C \\
 &= \frac{1}{2} \cdot \frac{y^{-4}}{-4} + C \\
 &= \frac{1}{2} \cdot -\frac{1}{4y^4} + C \\
 &= -\frac{1}{8y^4} + C
 \end{aligned}$$

2. Calculate $\int \frac{1}{3x^6} dx$

$$\begin{aligned}
 \int \frac{1}{3x^6} dx &= \frac{1}{3} \cdot \int x^{-6} \\
 &= \frac{1}{3} \cdot \frac{x^{-5}}{-5} + C \\
 &= \frac{1}{3} \cdot -\frac{1}{5x^5} + C \\
 &= -\frac{1}{15x^5} + C
 \end{aligned}$$

3. Calculate $\int \frac{\pi}{2} dz$

$$\begin{aligned}
 \int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\
 &= \frac{\pi}{2} \cdot z + C \\
 &= \frac{\pi z}{2} + C
 \end{aligned}$$

The sum rule for indefinite integrals

To take the integral of a sum, you can take the sum of the integrals of each term:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

The sum rule can be combined with the constant factor rule to integrate sums of constant multiples of functions:

$$\int (k_1 f(x) + k_2 g(x)) dx = k_1 \int f(x) dx + k_2 \int g(x) dx$$

Exercises

- Calculate $\int \sqrt{x}(2+x) dx$

$$\begin{aligned}
\int \sqrt{x}(2+x) \, dx &= \int 2\sqrt{x} + x \cdot \sqrt{x} \, dx \\
&= \int 2\sqrt{x} \, dx + \int x \cdot \sqrt{x} \, dx \\
&= 2 \int \sqrt{x} \, dx + \int x^{3/2} \, dx \\
&= 2 \cdot \frac{x^{3/2}}{\frac{3}{2}} + \frac{x^{5/2}}{\frac{5}{2}} + C \\
&= \frac{4x^{3/2}}{3} + \frac{2x^{5/2}}{5} + C \\
&= \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + C
\end{aligned}$$

- Calculate $\int \frac{2+x^4}{x^2} \, dx$

$$\begin{aligned}
\int \frac{2+x^4}{x^2} \, dx &= \int \frac{2}{x^2} \, dx + \int \frac{x^4}{x^2} \, dx \\
&= \int 2 \cdot x^{-2} \, dx + \int x^2 \, dx \\
&= 2 \cdot \int x^{-2} \, dx + \frac{x^3}{3} + C \\
&= 2 \cdot -x^{-1} + \frac{x^3}{3} + C
\end{aligned}$$

- Calculate:

$$\begin{aligned}
\int (3-x^2)^2 \, dx &= \int (9-6x^2+x^4) \, dx \\
&= 9 \int 1 \, dx - 6 \int x^2 \, dx + \int x^4 \, dx \\
&= 9x - 6 \frac{x^3}{3} + \frac{x^5}{5} + C \\
&= \frac{x^5}{5} - 2x^3 + 9x + C
\end{aligned}$$

- Find the antiderivative of $z^3 + 2z - 1$

$$\begin{aligned}
\int z^3 + 2z - 1 \, dz &= \int z^3 \, dz + \int 2z \, dz - \int 1 \, dz \\
&= \frac{z^4}{4} + 2 \frac{z^2}{2} - z + C \\
&= \frac{z^4}{4} + z^2 - z + C
\end{aligned}$$